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MEng Thesis Proposal

by

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Submitted to the Department of Electrical Engineering and Computer Science
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Abstract

This thesis surveys the cryptanalysis of the Learning Parity with Noise assumption and its variants. It focuses on the gap between the growing number of LPN-based assumptions and the relatively limited set of known reductions among them, especially when reductions are parameter-sensitive or restricted to particular regimes. The thesis will organize the main LPN variants, known reductions between search and decisional forms and across variants, classical attacks such as BKW and its descendants, and the current state of quantum cryptanalysis.

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1 Introduction

The Learning Parity with Noise assumption, usually abbreviated as LPN, is one of the central hardness assumptions connecting computational learning theory, coding theory, and modern cryptography. In its standard search form, the problem asks an adversary to recover a secret vector $s \in \mathbb{F}_2^n$ from noisy linear samples of the form

$$(a_i, b_i) = (a_i, \langle a_i, s\rangle \oplus e_i),$$

where each a_i is sampled from $\mathbb{F}_2^n$, each e_i is sampled from a noise distribution, and all arithmetic is over $\mathbb{F}_2$. Equivalently, for m samples one may write

$$b = As + e,$$

where $A \in \mathbb{F}_2^{m\times n}$, $b \in \mathbb{F}_2^m$, and e is a random error vector. The decisional version asks whether a collection of samples is drawn from this noisy-linear distribution or from the uniform distribution over $\mathbb{F}_2^n \times \mathbb{F}_2$. The simplicity of this formulation is one of LPN's main attractions: it is simple to state, efficient to instantiate, and closely related to the problem of decoding random linear codes. These properties make it especially appealing for lightweight cryptography and for settings where one wants post-quantum security.

LPN has been studied both as a learning problem and as a cryptographic assumption. Early work by Blum, Furst, Kearns, and Lipton showed how cryptographic primitives could be built from hard learning problems [1]. Later, Blum, Kalai, and Wasserman introduced the algorithm now known as BKW, a subexponential-time algorithm for learning parity functions in the presence of random noise [2]. This algorithm remains one of the foundational classical attacks on LPN and is still the starting point for much of the modern cryptanalytic literature.

LPN-based constructions have appeared in authentication protocols, symmetric-key primitives, commitments, zero-knowledge protocols, public-key encryption variants, and pseudorandom correlation generators [4, 5, 12]. LPN is also usually discussed as a post-quantum assumption, in the sense that no general-purpose quantum algorithm is currently known that breaks standard cryptographic parameter regimes in polynomial time. However, this should not be treated as a settled cryptanalytic fact. The classical landscape is already subtle, and the quantum landscape is comparatively less developed.

The modern use of LPN is no longer limited to the standard Bernoulli-noise problem over $\mathbb{F}_2$. In its standard search and decisional forms, LPN has been used for lightweight authentication, pseudorandom generation, and encryption [3, 5, 20]. Cryptographic applications also motivate a variety of related assumptions: low-noise and bounded-sample LPN for public-key encryption and sample amplification [6, 7], sparse-secret or sparse-parity LPN for somewhat homomorphic encryption [19], exact-noise and regular-noise LPN in PCG-style secure-computation settings

[18, 13], ring-LPN for efficient authentication and pseudorandom correlation generators [17, 18], and LPN over more general rings and fields for PCG applications over large fields and integer rings [13]. These variants often arise because a particular construction needs better efficiency, public-key functionality, structured algebra, or compatibility with secure computation protocols.

The motivating observation for this thesis is that the relationships among these variants are not as systematically understood as the growing use of the assumptions might suggest. Some reductions are known, including reductions between search and decisional formulations in standard settings, sample-amplification results for low-noise LPN, and recent reductions for LPN over integer rings and fields and between spare- and standard-LPN [4, 6, 13, 21, 22]. However, many relationships remain scattered across the literature, parameter-sensitive, or known only in specific regimes. This thesis will therefore be a survey and synthesis rather than a thesis centered on a new cryptanalytic discovery. Its goal is to organize the cryptanalysis of LPN and its variants into a coherent reference: first by defining the main variants, then by mapping known reductions, then by surveying classical algorithms, and finally by describing the current state of quantum cryptanalysis.

2 Related Work

The first body of related work concerns the origin of LPN as a cryptographic assumption. Blum, Furst, Kearns, and Lipton studied cryptographic primitives based on hard learning problems, providing early evidence that assumptions from learning theory could support cryptographic constructions [1]. The BKW algorithm of Blum, Kalai, and Wasserman later gave a subexponential-time approach to learning parity with noise [2]. In coding-theoretic language, LPN can be viewed as closely related to decoding random binary linear codes, which helps explain why both learning-theoretic and coding-theoretic attacks are relevant.

A second body of work concerns the use of LPN in cryptographic protocols. Authentication protocols such as HB and HB+ helped popularize LPN as a lightweight cryptographic assumption, especially for settings such as RFID devices where computation is limited [3, 4]. Pietrzak's survey gives a broader account of cryptography from LPN, including pseudorandom generators, symmetric-key encryption, authentication, commitments, and zero-knowledge proofs [5]. These works are important for this thesis because they explain why LPN is not merely a learning problem but a recurring assumption in efficient cryptographic design.

A third body of work concerns variants of LPN. Low-noise LPN appears in the context of public-key encryption and sample amplification. Döttling constructed public-key encryption with key-dependent message security based on low-noise LPN and studied sample amplification for this regime [6]. Kiltz, Masny, and Pietrzak used low-noise LPN in the construction of chosen-ciphertext secure public-key encryption [7]. These works show that changing the noise regime can enable different cryptographic applications, but they also raise questions about how low-noise assumptions relate to standard LPN assumptions.

Structured LPN variants are another major source of related work. Ring-LPN and related algebraic variants have become important in secure computation, especially in pseudorandom correlation generators. Boyle, Couteau, Gilboa, Ishai, Kohl, and Scholl constructed efficient pseudorandom correlation generators from ring-LPN [12]. Liu, Wang, Yang, and Yu recently studied the hardness of LPN over integer rings and fields for PCG applications, including relationships between search and decisional versions and reductions involving different noise formulations [13]. Bangachev, Bresler, Tiegel and Vaikuntanathan also recently studied the relationship between the hardness of standard and sparse-LPN, achieving the first known reductions between these variants paving a path forward on the formalization of the security of each version of the LPN assumption. [22]

The classical cryptanalysis of LPN is dominated by the BKW family of algorithms and its descendants. Levieil and Fouque improved the original BKW approach and applied their algorithmic analysis to LPN-based authentication parameters [8]. Bogos, Tramèr, and Vaudenay studied BKW and its variants from both theoretical and practical perspectives, including the sparse-secret setting [9]. Guo, Johansson, and Löndahl introduced a covering-code approach that improved attacks for many LPN parameter choices [10]. Esser, Kübler, and May also contributed to the practical cryptanalysis of LPN through improved decoding approaches [11]. These works provide the main material for the classical-algorithms portion of the thesis.

Finally, the quantum cryptanalysis of LPN is less mature than the classical literature. Generic quantum speedups, such as Grover search, can affect exhaustive-search components of some attacks [14]. More specifically, Sudo, Hara, Tezuka, and Yoshida initiated work on quantum search-to-decision reductions for LWE and also gave a search-to-decision reduction for LPN [15]. Recent exploratory work has also investigated quantum approaches to LPN more directly [16]. These works suggest that quantum cryptanalysis of LPN is an active and developing area, but not one in which there is yet a single dominant analogue of the BKW classical framework.

3 Proposed Work

This thesis will produce a structured survey of the cryptanalysis of LPN and its variants. The central contribution will be organizational and comparative: it will collect definitions, reductions, attacks, and open questions into a coherent map. The intended reader is someone familiar with basic cryptography and linear algebra but not necessarily familiar with the full LPN literature.

The first part of the thesis will define the main LPN variants. This section will begin with standard search and decisional LPN over $\mathbb{F}_2$, then describe how variants arise by changing the sample distribution, secret distribution, noise distribution, number of available samples, or algebraic domain. The variants to be covered will include low-noise LPN, bounded-sample LPN, exact-noise LPN, regular-noise LPN, sparse-secret LPN, ring-LPN, and LPN over integer rings and fields. For each variant, the thesis will state the sample distribution, the adversarial goal, the main parameters, and the cryptographic motivation. One planned deliverable is a taxonomy table comparing the variants by domain, secret distribution, noise model, sample regime, and known applications.

The second part will study reductions between variants. This will be one of the central sections of the thesis. The goal is to create a reduction map showing which variants are known to imply which others, under what parameter losses, and in which direction. This section will include search-to-decision reductions, reductions involving low-noise and bounded-sample LPN, and recent results for LPN over integer rings and fields [4, 6, 13]. A major theme will be that reductions should not be summarized only as "known" or "unknown." In cryptographic applications, it matters whether a reduction is dimension-preserving, sample-preserving, noise-preserving, tight enough for concrete security, or only asymptotically meaningful. The thesis will therefore distinguish between qualitative equivalence and parameter-preserving equivalence.

The third part will survey classical algorithms for LPN. This section will devote substantial attention to the BKW algorithm. It will explain the reduction phase, in which samples are combined to eliminate blocks of secret coordinates, and the solving phase, in which the remaining lower-dimensional problem is attacked statistically. It will then discuss how later work modifies this framework, including the Levieil–Fouque improvement, BKW variants analyzed by Bogos, Tramèr, and Vaudenay, covering-code approaches, and later decoding-based improvements [2, 8, 9, 10, 11]. This section will also compare dense-secret and sparse-secret regimes, since sparse-secret LPN can favor attacks that are not simply direct applications of the standard BKW method.

The fourth part will survey the quantum cryptanalysis landscape. The purpose of this section will not be to claim that known quantum algorithms break LPN, but to identify what quantum computation currently changes and what remains open. The section will separate generic quantum speedups, such as Grover-style improvements to exhaustive-search components, from more LPN-specific results such as quantum search-to-decision reductions and exploratory quantum approaches to sample generation or transformation [14, 15, 16]. Because some of this literature is recent and less established than the classical BKW literature, the thesis will distinguish peer-reviewed results from preprints and speculative directions.

The final part of the thesis will synthesize the survey into a conclusion and further-work section. This section will identify gaps in the current literature, including missing reductions between commonly used variants, parameter losses that make some reductions less useful for concrete security, and the lack of a systematic quantum cryptanalytic framework for structured LPN assumptions. The thesis will also propose directions for future work, such as a standardized notation for LPN variants, a public reduction graph, concrete-security comparison tables, and more systematic treatment of ring and integer-ring assumptions.

The expected output of this thesis is not a new algorithm or a new reduction theorem, although such observations may be included if they arise naturally during the survey. The required output will be a technically accurate and carefully cited survey that clarifies the current state of knowledge. In particular, the thesis will produce three concrete artifacts: a taxonomy of LPN variants, a map of known reductions and missing relationships, and a comparative survey of classical and quantum attacks.

4 Proposed Timeline

Date	**Milestone**
May 15, 2026	Thesis scope, title, chapter list, and research questions frozen.
May 29, 2026	Core bibliography assembled, but not fully annotated.
June 19, 2026	Preliminaries and taxonomy chapters drafted.
July 3, 2026	Classical algorithms and variant-sensitive algorithms drafted.
July 17, 2026	Reductions and quantum algorithms chapters drafted.
July 24, 2026	Constructions and holistic-framework chapters drafted.
July 31, 2026	Complete first draft sent to advisor.
August 14, 2026	Revised second draft completed.
August 21, 2026	Near-final version sent to advisor/readers.
August 28, 2026	Final revision pass complete.
September 4, 2026	Submission-ready thesis completed.

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